

The Lach Elemental Lattice

An Order-Theoretic Account of the Periodic System

Construction, Order Structure, and an Exploratory Application

in four parts

- I · The Configuration-Space Lattice
- II · The Lattice as an Ordered Set
- III · The Periodic System within the Lattice
- IV · An Exploratory Application to Deuteride Host Selection

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Abstract

Part I develops a three-axis lattice in which every element occupies a coordinate (n, ℓ, k) defined by the quantum numbers of its ground-state valence configuration. All 118 known elements are placed; the surrounding empty volume is classified as void ($\ell \geq n$, orbital non-existent), reactive (k exceeds Pauli capacity), or reserved (valid but unoccupied), giving 180, 240 and 92 cells respectively against 118 occupied. The displacement vector ΔQ sorts the elements into three geometric classes, and the lattice makes visible an adjacency the conventional table conceals: the late f-block and the late 5d metals occupy neighbouring regions of configuration space, whereas the standard layout exiles the lanthanides to a detached strip. The coordinates are then tested against measured data and benchmarked against prior art. Regressing hydride formation enthalpy for 39 metals gives a two-term model in ℓ and k with cross-validated $R^2 = 0.51$ — not competitive, since Pettifor's Mendelev number reaches 0.79 using a single variable and Miedema's model 0.90. One result survives: the occupancy coordinate k is uncorrelated with every established descriptor tested, and appending it to Pettifor's scale raises cross-validated R^2 from 0.79 to 0.84 (permutation $p = 0.001$).

Part II turns from prediction to structure, treating Λ as an abstract ordered set. It begins by asking whether three coordinates are necessary: exhibiting a standard example S_3 among the admissible cells shows that the order dimension is exactly three, so no pair of linear orders can encode the containment relation. Exhaustive verification then establishes that the admissible cells form a bounded lattice, with hydrogen at $(1, 0, 1)$ as the unique minimum element rather than a chosen anchor; counting cells at fixed n returns the Pauli capacities $2n^2$ exactly. The lattice is Sperner though not rank-symmetric, and its rank generating function decomposes into shifted q -integers and vanishes at $q = -1$, an exact parity balance forced by spin degeneracy. The Madelung rule appears as a linear functional $n + \ell$ whose level sets are parallel planes, and the volumes of those slices pair exactly — 2, 2, 8, 8, 18, 18, 32, 32 — reproducing the

observed period lengths, which the conventional layout must impose rather than explain. Projecting along this functional yields the Janet left-step table exactly. Fixing (ℓ, k) while varying n instead yields fibres reproducing the conventional chemical groups, with one exception: helium falls with the alkaline earths, the Janet placement reached by a second and independent route. The kainosymmetric subshells $1s$, $2p$, $3d$ and $4f$ are exactly the boundary diagonal $\ell = n - 1$. A standard sublattice argument explains the closure and shows why the hydrogenic labelling (n, ℓ, m, s) fails, the four-axis unfolding proposed in an earlier draft being withdrawn.

Part III asks where the periodic system sits within that structure. The 118 occupied cells form an order ideal of Λ , closed downward without exception, which follows from aufbau filling being a monotone accumulation. The ideal is moreover all-or-nothing by column: of the twenty-five (n, ℓ) columns, nineteen are filled to Pauli capacity and six are empty, none partially, so the realised system is one of only 120 down-sets of the column poset, at the 79th percentile by size. Its three maximal cells are those of lawrencium, copernicium and oganesson, leaving five covering addresses for any further element — a constraint on valence configuration alone, not on nuclear stability. The relation of this framework to existing order-theoretic treatments of chemistry is set out, and novelty is not claimed for the techniques employed.

Part IV applies the adjacency of Part I to deuteride host selection and reports the attempt honestly, including its failure. A heuristic index combined with a crystal-structure filter screens 118 elements to nine candidates, with ytterbium the notable entry: an FCC hydride-former apparently untested in the loading literature. But the ordering is shown to depend on which of several defensible conventions is adopted, and the framework yields dimensionless ratios that cannot be converted into tunnelling rates without an unconstrained external scale — the axes being counts, so that $\Lambda \subseteq \mathbb{N}^3$ makes the obstruction intrinsic. Part IV is offered as a candidate-generation heuristic, not a theory of low-energy nuclear reactions.

PART I — THE CONFIGURATION-SPACE LATTICE

1. Introduction

Mendeleev's table remains the most productive organising device in chemistry, and its two-dimensional form encodes real periodicity: rows track the principal quantum number, columns track valence behaviour. But the layout imposes compromises that are cartographic rather than physical. The f-block is removed from the body of the table and printed below it, for reasons of page width alone. Subshell type ℓ and occupancy k are conflated into a single positional axis, so two elements in one group may share electron count while occupying different subshells. And the table has no vocabulary for absence: a gap either holds an undiscovered element or is nothing at all, with no way to say which vacancies are physically possible and which are forbidden.

This paper treats n , ℓ and k as three independent axes and asks what becomes visible. The construction is deliberately minimal — no new physics is introduced, only a change of coordinates. What the change buys is set out in four parts. Part I constructs the lattice and reports what it makes visible — a classification of empty space, a geometric sorting of the elements into three classes, and an adjacency relation that the conventional layout destroys — then tests the coordinates against measured hydride thermodynamics and benchmarks them against established descriptors. Part II treats Λ as an abstract ordered set and contains the results of which this work can be most confident. Part III asks where the 118 known elements sit within that structure. Part IV applies the framework to a concrete materials question and reports, at length, why the attempt does not succeed.

2. Construction of the Lattice

Each element is assigned the coordinate (n, ℓ, k) of its ground-state valence configuration, where n is the principal quantum number, $\ell \in \{0,1,2,3,4\}$ denotes s, p, d, f, g, and k counts the electrons occupying that subshell. The admissible lattice is

$$\Lambda = \{ (n, \ell, k) \in \mathbb{Z}^3 : 0 \leq \ell \leq n-1, \quad 1 \leq k \leq 2(2\ell+1) \} \quad (1)$$

the two constraints being the angular-momentum condition and the Pauli capacity of the subshell. Hydrogen anchors the origin at $(1, 0, 1)$. A fifth layer at $\ell = 4$ is carried as a theoretical extension; no known ground state occupies it.

Two features of this construction deserve emphasis, because they will matter in Part II. First, the axes are quantum numbers, not lengths — n indexes a shell, ℓ indexes angular momentum, k is a count. Distances in the lattice are therefore combinatorial, not metric. Second, the axes are coupled: the range of ℓ depends on the value of n , and the range of k depends on ℓ . The lattice is a bounded, interdependent integer structure, and this is precisely what allows the classification in Section 3.

3. The Classification of Empty Space

Every cell not occupied by a known element falls into exactly one of three regimes, and the distinction is the first substantive product of the coordinate change.

Void cells satisfy $\ell \geq n$. The orbital does not exist for that shell, and no element can ever occupy the cell under any conditions. Reactive cells satisfy $\ell < n$ but $k > 2(2\ell+1)$: the electron count exceeds Pauli capacity, so the configuration cannot be a ground state; under the reading proposed here these correspond to transient overflow states in which the excess electron is promoted to a new shell. Reserved cells satisfy

both constraints yet hold no known element. These are valid, physically permitted configurations that are simply unoccupied.

A full census of the $7 \times 5 \times 18$ lattice is given in Figure 1. Of 630 cells, 180 are void, 240 reactive, 92 reserved, and 118 occupied. The reserved count corrects the figure of 78 quoted in the earlier draft of this work; the recount is exact and is reproduced by the classification code accompanying this paper.

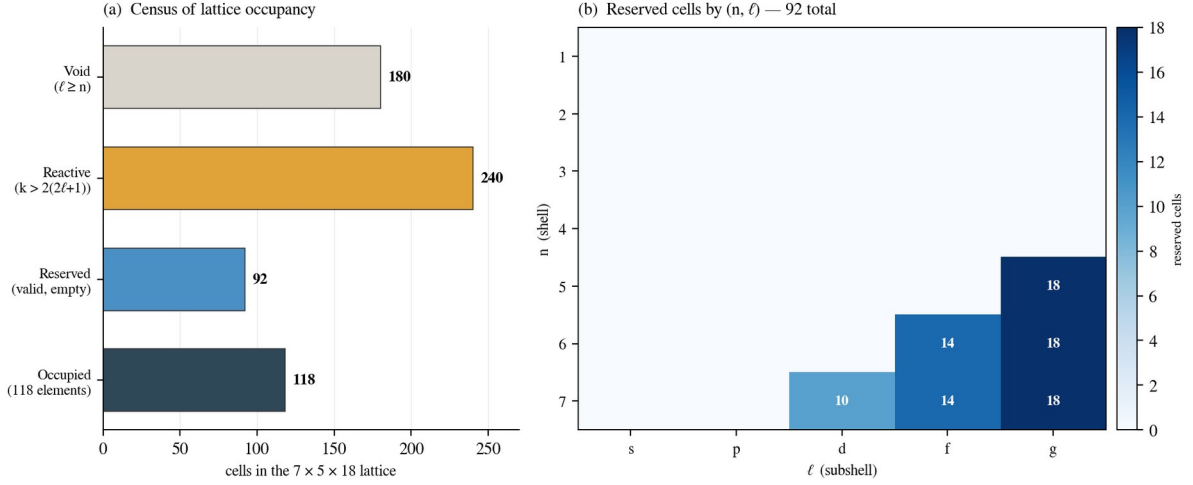


Figure 1. Census of the configuration-space lattice. (a) The 630 cells partition into four regimes, with the void and reactive classes together accounting for two-thirds of the volume. (b) Distribution of the 92 reserved cells across (n, ℓ) . Reserved cells concentrate at $n = 6-7$ in the d, f and g layers — the region corresponding to elements beyond $Z = 118$.

The concentration visible in Figure 1(b) is the point. Reserved cells are not scattered uniformly through the empty volume; 42 of the 92 sit at $n = 7$ and a further 32 at $n = 6$, overwhelmingly in the higher- ℓ layers. In the two-dimensional table this region is simply the blank space past oganesson. Here it is a set of addressed coordinates, each with a defined neighbourhood.

4. The Elements in Place

Figure 2 shows all 118 elements at their ground-state coordinates. No manual grouping is applied: the block structure emerges from the ℓ -axis alone, and within each block the elements stratify along n by period. The f-block, which the conventional table detaches entirely, sits in the body of the structure at $\ell = 3$.

(a) oblique view

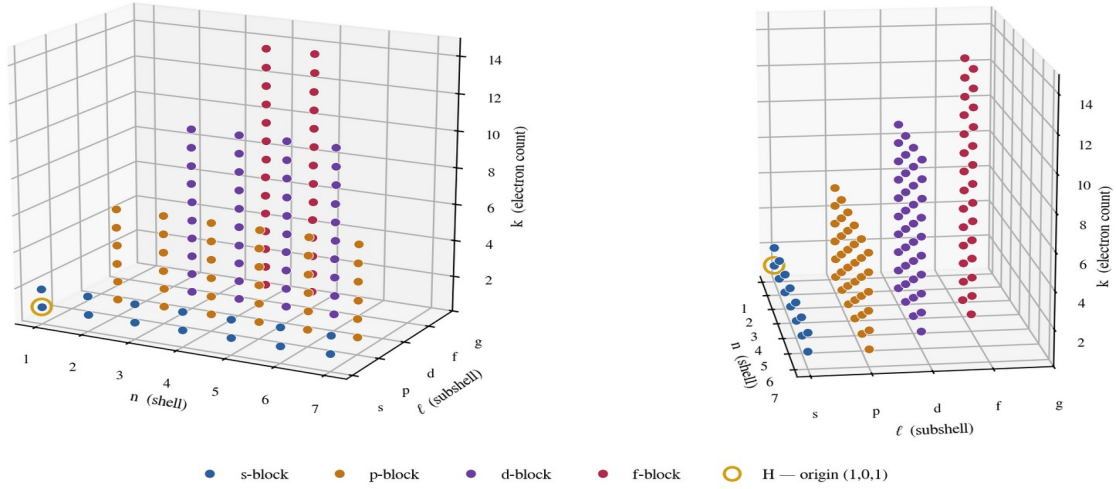
(b) near-edge view — block separation along ℓ 

Figure 2. The 118 known elements at their ground-state coordinates (n, ℓ, k) . Colour denotes block; the gold ring marks hydrogen at the origin $(1, 0, 1)$. (a) Oblique view. (b) Near-edge view showing the clean separation of blocks along the ℓ -axis. The clustering is a consequence of the coordinates, not of any imposed arrangement.

5. Excited Neighbourhoods and the Field Equation

Each element is modelled as a three-state system: its ground configuration $Q_0 = (n, \ell, k)$ together with its two nearest excited neighbours,

$$Q_1 = (n+1, \ell, k) \quad [\text{shell excitation}] \quad (2)$$

$$Q_2 = (n, \ell \pm 1, k) \quad [\text{subshell excitation}] \quad (3)$$

with Q_2 taking $\ell+1$ where the angular-momentum condition permits and $\ell-1$ otherwise. Writing the state as a superposition over the accessible cells and requiring normalisation at all t , ground-state occupancy at $t = 0$, and sinusoidal evolution at base frequency Ω gives coefficients $c_0 = \cos(\Omega t)$ and $c_1 = c_2 = \sin(\Omega t)/\sqrt{2}$. The expectation position $\langle Q \rangle(t)$ then traces a bounded path through configuration space, and the variance $\sigma^2(t)$ measures the instantaneous spread. Phase-shifting the coefficients by $\pi/2$ yields the inverse (emission) expression, and combining the two gives a unified form, designated here the Lach elemental field equation

$$\Psi_{\pm Z}(t) = \sqrt{2} \cdot \sin(\Omega t + \pi/4) \cdot \sum_{(n, \ell, k) \in \Lambda} |n, \ell, k\rangle \quad (4)$$

The displacement from the midpoint of the excited pair back to the ground state,

$$\Delta Q = Q_0 - \frac{1}{2}(Q_1 + Q_2) \quad (5)$$

is a static vector characterising the geometry of each element's neighbourhood, and the interaction term $S_Z(t) = \cos(2\Omega t) \cdot \Delta Q$ oscillates at twice the base frequency.

A caution on interpretation. Equation (4) is a construction on the lattice, not a solution of a Hamiltonian derived from physical potentials. Ω is a formal parameter; nothing in this paper establishes a correspondence between Ω and a measurable frequency in any material. The expression is also element-independent: the summation carries no element label and the prefactor is common to all, so every element shares a single envelope. What distinguishes one element from another is not Equation (4) but the displacement vector ΔQ and the spread $\sigma^2(t)$, which are geometric quantities read off the lattice. Section

12.4 returns to this point; the results that follow depend on the geometry of ΔQ and not on any dynamical claim.

6. Three Geometric Classes

Evaluating ΔQ across all 118 elements yields a sharply degenerate result. The magnitude $|\Delta Q|$ takes exactly two values: unity for hydrogen and helium, and $1/\sqrt{2}$ for every other element. The elements are differentiated not by the size of the displacement but by its direction, which falls into three cases.

Class A comprises hydrogen and helium alone. Here $n = 1$ admits only $\ell = 0$, so both excited neighbours collapse onto the same cell, $Q_1 = Q_2$. The neighbourhood is one-dimensional and the spread at maximum excitation vanishes.

Class B comprises elements for which $\ell + 1 \leq n - 1$, so subshell excitation moves toward higher angular momentum. Class C comprises elements for which only $\ell - 1$ is available, so subshell excitation moves toward lower angular momentum. Figure 3 shows the membership.

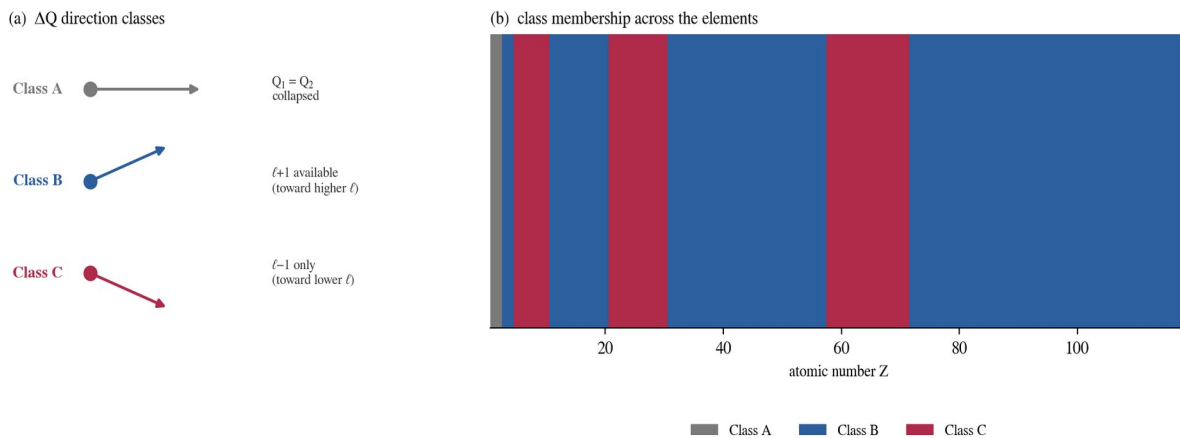


Figure 3. Direction classes of the displacement vector ΔQ . (a) Schematic of the three cases. (b) Class membership across atomic number. The 3d series and the lanthanides fall in Class C; the 4d and 5d series fall in Class B. The classification follows entirely from the angular-momentum constraint $\ell \leq n-1$ and requires no additional input.

The classification is not arbitrary bookkeeping. It states which direction in configuration space each element's accessible excitations lie, and it separates the 3d and 4f series from the 4d and 5d series on purely geometric grounds.

7. The Adjacency Result

The most consequential structural finding is illustrated in Figure 4. Consider ytterbium, (4, 3, 13), and platinum, (5, 2, 8). In the conventional table these elements are maximally separated by the layout: Yb sits in the detached lanthanide strip printed beneath the body of the table, Pt sits in period 6 of the main block. Nothing in the printed arrangement suggests any relationship between them.

In the lattice they are close. Both occupy the high- k region of the high- ℓ layers, separated by one step in ℓ and a few in k , and both sit at large configuration-space distance from the origin — $|Q_0| = 13.93$ for Yb and 9.64 for Pt, against 9.17 for palladium. The late f-block and the late 5d metals form a contiguous region that the two-dimensional layout tears in half.

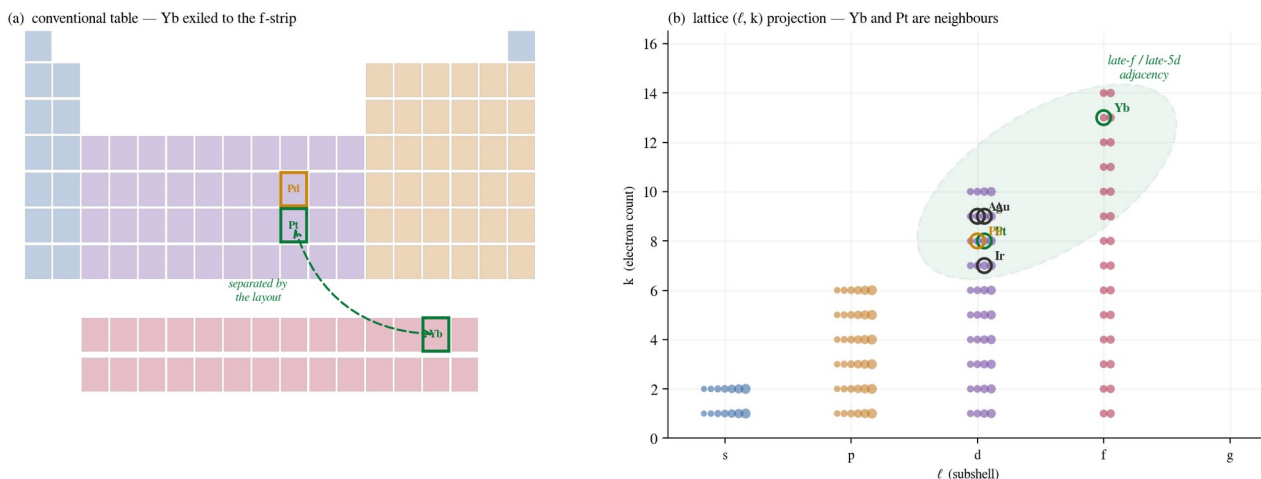


Figure 4. The adjacency the conventional layout conceals. (a) In the standard table, ytterbium is displaced into the lanthanide strip, far from platinum; the dashed arc indicates the separation imposed by the layout. (b) In the (ℓ, k) projection of the lattice, Yb and Pt are neighbours within a contiguous high- $|Q_0|$ region. Marker size encodes n . This adjacency is the principal structural result of Part I and the point of departure for Part II.

8. Testing the Coordinates Against Measured Data

Sections 3 to 7 establish structural facts about the lattice. None of them involves a measurement. It is reasonable to ask whether the coordinates carry information about physical quantities, and equally reasonable to ask how any such information compares with what established descriptors already provide. This section reports both tests. The second is the more important, and it is unflattering.

8.1 The regression

The quantity chosen is the enthalpy of hydride formation, ΔH_f , in kJ per mole of hydrogen: measured, tabulated, dimensional, independent of this framework, and the quantity that governs whether a metal absorbs hydrogen and how strongly it binds it. Values were taken for 39 transition metals, lanthanides and actinides with well-established binary hydride data. Ordinary least squares on the lattice coordinates gives

$$\Delta H_f = +136.2 - 89.5 \cdot \ell + 6.4 \cdot k \quad (\text{kJ} \cdot \text{mol}^{-1} \text{ H}) \quad (7)$$

with both coefficients significant — ℓ at $p = 3.1 \times 10^{-8}$, k at $p = 8.1 \times 10^{-4}$ — an in-sample R^2 of 0.592, and a leave-one-out cross-validated R^2 of 0.514. Adding n , $|Q_0|$ or H_{cp} does not improve the cross-validated fit; the five-term model gains in-sample R^2 while losing it out of sample, the signature of overfitting. Atomic number alone gives a negative cross-validated R^2 , carrying no predictive information on this data. Figure 5 presents the fit.

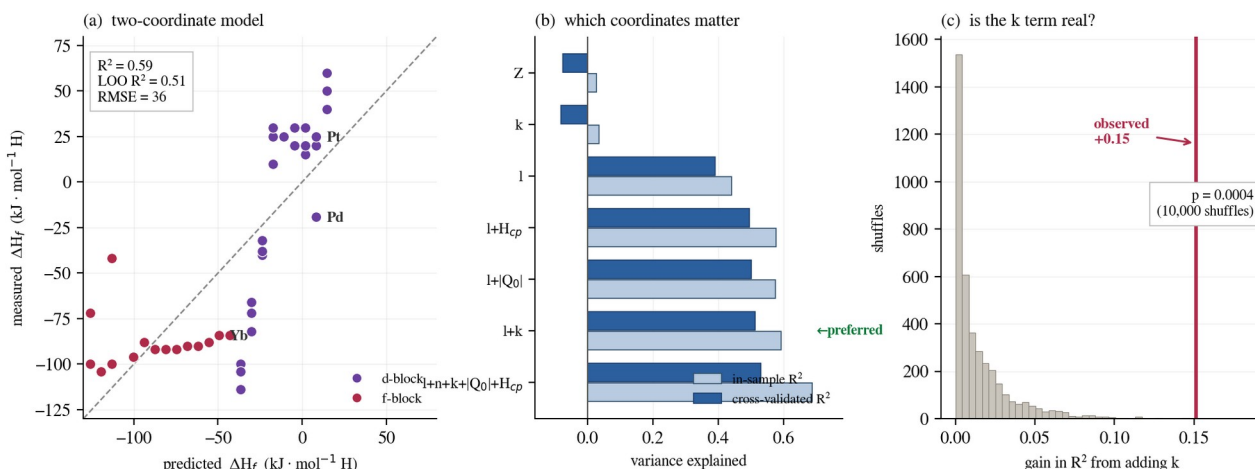


Figure 5. The lattice coordinates against hydride formation enthalpy for 39 metals. (a) Measured versus predicted for the model of Equation (7); the dashed line is parity. (b) Variance explained by candidate models, in-sample and cross-validated. (c) Permutation null for the contribution of k : the observed gain of 0.15 in R^2 lies outside the distribution from 10,000 shuffles ($p = 0.0004$).

Taken alone this looks encouraging. It should not be taken alone.

8.2 Benchmarked against established descriptors

Two schemes address this problem and have done so for decades. Pettifor's Mendelev number is a one-dimensional chemical scale constructed so that structure maps separate cleanly; Miedema's model predicts formation enthalpies from an electronegativity parameter φ^* and an electron-density term $n_{ws}^{1/3}$. Both were fitted to the same 39 elements under identical cross-validation. The comparison is given in Figure 6 and Table 1.

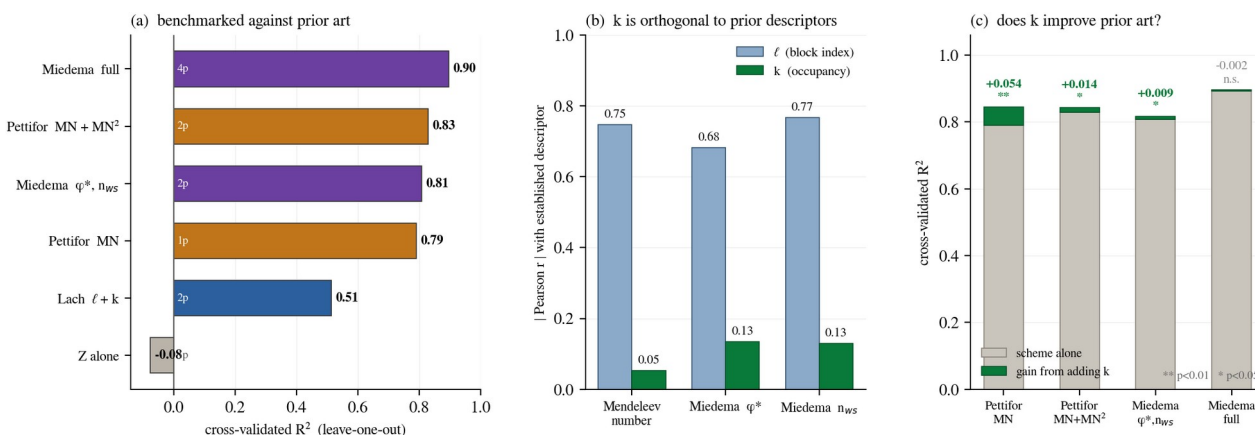


Figure 6. The lattice weighed against prior art. (a) Cross-validated R^2 for each scheme, with the number of fitted predictors marked. Pettifor's single-variable scale outperforms the two-variable lattice model by a wide margin, and Miedema's full form outperforms both. (b) Absolute correlation of the lattice coordinates with the established descriptors. The block index ℓ is strongly correlated with all three — it encodes information they already carry — whereas the occupancy k is uncorrelated with all three. (c) Cross-validated R^2 of each established scheme before and after adding k , with permutation significance. The occupancy coordinate improves Pettifor's scale and the two-parameter Miedema form, but adds nothing to the full Miedema model.

Table 1. Prediction of hydride formation enthalpy across 39 metals, all schemes fitted and cross-validated identically. Paired Wilcoxon tests compare squared leave-one-out errors against the lattice model.

Scheme	Predictors	Fit R^2	CV R^2	CV RMSE	vs. lattice
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Miedema, full form	4	0.934	0.895	18.1	$p < 0.0001$
Pettifor, $MN + MN^2$	2	0.854	0.828	23.2	$p < 0.0001$
Miedema, $\varphi^* + n_{ws}$	2	0.842	0.808	24.6	$p = 0.0001$
Pettifor, MN alone	1	0.808	0.790	25.7	$p = 0.0003$
Lattice, $\ell + n + k$	3	0.608	0.504	39.4	—
Lattice, $\ell + k$	2	0.592	0.514	39.1	—
Atomic number Z	1	0.028	-0.076	58.1	$p < 0.0001$

The result is unambiguous and runs against the framework. Pettifor's scale, using a single variable, achieves a cross-validated R^2 of 0.790 against the lattice's 0.514 with two. Miedema's full form reaches 0.895. Paired Wilcoxon tests on the leave-one-out squared errors reject equality against every established scheme, the weakest at $p = 0.0003$. On the task of predicting hydride formation enthalpy, the lattice coordinates are substantially outperformed by prior art that has been available since the 1980s.

Nor does combining them rescue the position. Adding ℓ and k to the full Miedema model changes its cross-validated R^2 from 0.895 to 0.888 — a decrease. Whatever the lattice contributes on this task is already contained in the established descriptors.

8.3 What survives: the occupancy coordinate

One result does survive the benchmark, and it is worth isolating precisely because the rest did not.

Figure 6(b) shows that the two lattice coordinates behave quite differently. The block index ℓ correlates strongly with every established descriptor — $|r|$ between 0.68 and 0.77 — which is expected, since ℓ is the block label and every chemical scale encodes block membership. The occupancy k , by contrast, is uncorrelated with all three: $r = -0.05$ against the Mendelev number, $+0.14$ against φ^* , -0.13 against $n_{ws}^{1/3}$, none approaching significance. The occupancy coordinate is orthogonal to the established descriptors.

Figure 6(c) tests whether that orthogonal information is useful. Adding k to Pettifor's Mendelev number raises cross-validated R^2 from 0.790 to 0.844, a gain of 0.054 significant at $p = 0.001$ by permutation. Smaller but significant gains follow for the quadratic Pettifor form ($+0.014$, $p = 0.029$) and the two-parameter Miedema form ($+0.009$, $p = 0.043$). Against the full four-parameter Miedema model the gain vanishes (-0.002 , $p = 0.363$), which is unsurprising: with four fitted per-element parameters that model has already absorbed the available variance.

The defensible claim is therefore narrow and specific. Valence subshell occupancy carries information about hydride stability that Pettifor's chemical scale does not encode, and appending it to that scale improves prediction measurably. This is a modest contribution to an existing method rather than a competitive alternative to it, and it is stated here in those terms.

Two further points. The classification by ΔQ direction from Section 6, derived purely from the angular-momentum constraint, separates the same data: Class C elements average $-61.9 \text{ kJ} \cdot \text{mol}^{-1} \text{ H}$ against -15.7 for Class B, significant at $p = 0.010$ by Welch's test. And the ℓ term in Equation (7) is not a discovery — that f-block metals form more stable hydrides than late d-block metals is long-established descriptive chemistry, which the lattice recovers rather than reveals.

It is equally important to state the limit. Everything in this section concerns bonding energetics of order 10^{-1} eV over interatomic distances of order 10^{-10} m . It establishes that the lattice has dimensional content in the chemical regime, empirically and by regression against measured values, in the same tradition as the schemes it was benchmarked against. It carries no implication for quantities at nuclear scale, and Section 19 returns to this.

9. Reserved Cells and Undiscovered Elements

The 92 reserved cells are addressed vacancies: each carries coordinates, a defined excited neighbourhood via Equations (2) and (3), and hence a computable ΔQ . Their concentration at $n = 6-7$ in the d, f and g layers corresponds to the region beyond $Z = 118$, and the framework therefore offers a geometric description of where a superheavy element would sit and what its configuration-space neighbourhood would look like.

It is worth being precise about what this does and does not amount to. The lattice states that certain configurations are permitted and unoccupied. It does not predict nuclear stability, synthesis cross-sections, or half-lives, all of which are governed by nuclear structure rather than valence configuration. The reserved-cell analysis constrains where to look in configuration space; it says nothing about whether a nucleus placed there would survive long enough to be observed. Claims in the earlier draft that the framework transforms the search for new elements into a bounded problem overstate this, and are withdrawn.

PART II — THE LATTICE AS AN ORDERED SET

Prefatory note. Part I asked what the lattice contains and how well it predicts. This part asks what kind of object the lattice is, treating Λ as an abstract ordered set without reference to which of its cells are occupied. The results are of a different character from those of Part I: they are structural facts about the coordinate system rather than correlations with data, they do not depend on the origin or on any dynamical reading of Equation (4), and several of them explain why the construction works at all. Part III then returns to the elements, asking where the periodic system sits within the structure established here.

10. Λ as a Bounded Algebraic Lattice

10.1 *The order dimension of Λ is three*

Part I assigned three coordinates to each element without arguing that three were necessary. Two would be simpler and the conventional table manages with two axes; the question is whether the third carries information that cannot be recovered otherwise. Order dimension answers it precisely.

The order dimension of a poset, introduced by Dushnik and Miller, is the least number of linear orders whose intersection reproduces the partial order. A poset of dimension two can be encoded by two rankings: x precedes y in the partial order exactly when it precedes y in both. Dimension three means no such pair of rankings exists.

Since Λ is a subset of $\mathbb{N}^3$ ordered componentwise, and the three coordinate projections are linear orders whose intersection is precisely the componentwise order, the dimension is at most three. To establish that it is exactly three it suffices to exhibit a standard example S_3 : three elements a_1, a_2, a_3 and three elements b_1, b_2, b_3 with $a_i \leq b_j$ if and only if $i \neq j$. Such a configuration cannot be realised by two linear orders, so its presence forces dimension at least three.

The cells

$$\begin{aligned} a_1 &= (7, 2, 6) & a_2 &= (5, 4, 6) & a_3 &= (5, 2, 8) \\ b_1 &= (5, 4, 8) & b_2 &= (7, 2, 8) & b_3 &= (7, 4, 6) \end{aligned} \quad (9)$$

are all admissible, and each a_i lies below exactly those b_j with $j \neq i$, as Figure 7 shows. Each pair differs from the base cell $(5, 2, 6)$ by an increment of two in one coordinate for the minimals and in two coordinates for the maximals, which is what produces the cyclic incidence pattern. Hence $\dim(\Lambda) = 3$ exactly.

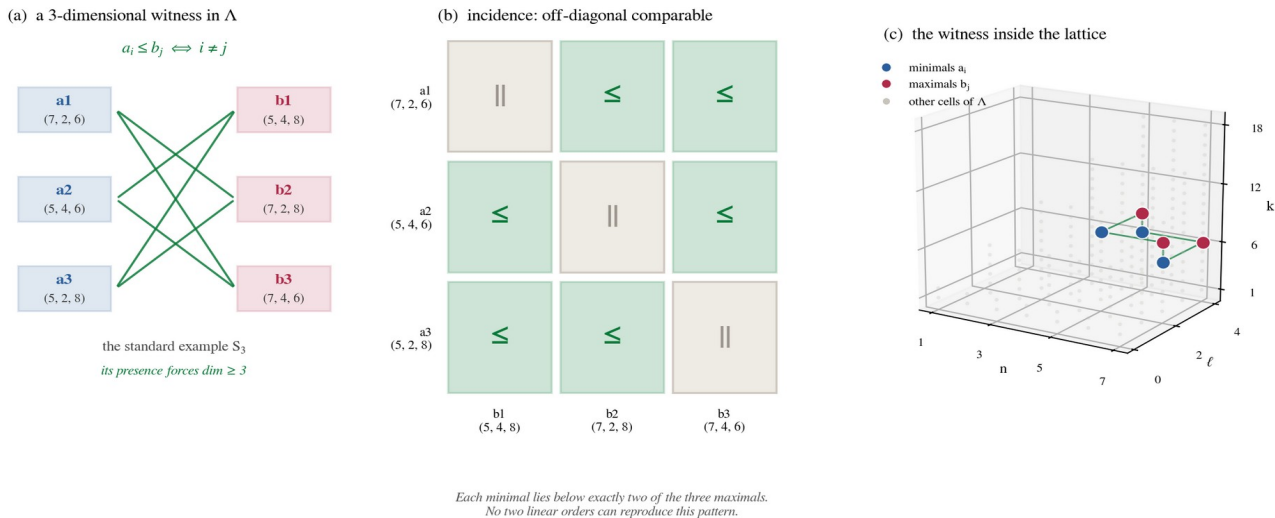


Figure 7. Order dimension of the lattice. (a) Six admissible cells forming a standard example S_3 : each minimal a_i lies below exactly two of the three maximals, omitting b_i . (b) The incidence pattern, comparable off the diagonal and incomparable on it. A pair of linear orders cannot reproduce it. (c) The witness located within Λ , other admissible cells shown faintly. Since $\Lambda \subseteq \mathbb{N}^3$ gives $\dim \leq 3$ and the witness gives $\dim \geq 3$, the order dimension is exactly three.

The consequence is worth stating carefully, because the result is easy to overread. What it establishes is that no two linear orders can encode the containment relation on Λ : any two-coordinate scheme, however contrived, must lose order information. In that specific sense the three coordinates are necessary rather than merely convenient, and the periodic system is irreducibly three-dimensional as an ordered set.

What it does not establish is anything about physical space. Order dimension is a property of a relation, not a spatial claim, and the axes remain counts of nodes and electrons as Section 19 emphasises. Nor does it imply that two-dimensional periodic tables are deficient for their purposes: a layout may usefully display chemical similarity without encoding the componentwise order at all, and the conventional table does exactly that. The claim is confined to the order relation defined in Section 2.

The witness given here occupies cells at $\ell = 4$, the g-subshell, which no known element reaches. This is legitimate for a statement about Λ , whose admissible region includes those cells by construction, but it means the result concerns the lattice rather than the occupied set alone. Restricting attention to the 118 occupied cells would require a separate argument, which is not attempted here.

The technique is standard — the standard example and the Dushnik–Miller characterisation are textbook material, and the argument above is an application rather than a development. No search of the order-theoretic literature has been undertaken beyond the works discussed in Section 14, so novelty is not claimed; the result is offered because it justifies the construction of Part I, not because it is believed to be new.

10.2 Closure under join and meet

Order the admissible cells componentwise: $(n, \ell, k) \leq (n', \ell', k')$ when each coordinate is no greater than its counterpart. The question is whether Λ is closed under the resulting join and meet — that is, whether the componentwise maximum and minimum of any two admissible cells are themselves admissible. Exhaustive checking of all 21,945 pairs drawn from the 210 admissible cells returns no failures in either direction.

The reason is structural. Raising n never violates $\ell \leq n - 1$, and raising ℓ raises the capacity ceiling $2(2\ell+1)$, so a raised k remains admissible; lowering ℓ lowers the ceiling but k is lowered in step. Both constraints are therefore preserved under join and meet, and Λ is a bounded lattice with least element $(1, 0, 1)$ and greatest element $(7, 4, 18)$.

One consequence deserves emphasis. Hydrogen was placed at the origin in Section 2 as a reference choice. It is not a choice: $(1, 0, 1)$ is the unique minimum of the poset, the only cell below every other. The anchoring of the framework at hydrogen is a theorem about the structure rather than a convention imposed on it.

A related point settles a question that might otherwise seem open. The quantities ΔQ and $\sigma^2(t)$ are differences of coordinates and are therefore invariant under any translation of the origin; so are the direction classes of Section 6 and the adjacency of Section 7. Order-theoretic facts are likewise preserved, since a coordinate shift is an order isomorphism. Only $|Q_0|$, and hence H_{cp} , depends on where the origin is placed — and that quantity was shown in Section 8 to have negative cross-validated predictive value under every origin tested. Nothing that works in this framework depends on the origin, and nothing that fails is rescued by moving it.

10.3 Shell capacity as lattice volume

Counting the admissible cells at fixed n gives

$$\sum_{(\ell=0 \text{ to } n-1)} 2(2\ell+1) = 2n^2 \quad (8)$$

so the shell capacities 2, 8, 18, 32, 50, 72, 98 are the volumes of the horizontal slices of Λ . The Pauli capacities of the periodic system are not an input to the lattice; they are what one obtains by counting it. This is a small result, but it indicates that the coordinate choice is capturing genuine structure rather than merely relabelling it.

10.4 Two further invariants

Two properties of Λ are worth recording, both established by direct computation and both bearing on how the lattice compares with familiar combinatorial structures.

The first is the Sperner property. A graded poset has this property when no antichain exceeds the largest rank level in size. Products of chains possess it automatically, being rank-symmetric and unimodal, but Λ is not rank-symmetric, as Section 13.5 shows, so the property cannot be inherited and must be checked. Computing the maximum antichain by Dilworth's theorem, via a maximum matching in the comparability graph, returns 15 — exactly the size of the largest rank level, at $n + \ell + k = 11$. Λ is Sperner.

The second is a parity invariant. The rank generating function of Λ decomposes as a sum of shifted q -integers, one per column,

$$F(q) = \sum_{(n,\ell)} q^{(n+\ell+1)} \cdot [2(2\ell+1)]_q, \quad [m]_q = 1 + q + \dots + q^{(m-1)} \quad (10)$$

an identity verified symbolically. Every column capacity $2(2\ell+1)$ is even, and $[m]_q$ vanishes at $q = -1$ whenever m is even, so $F(-1) = 0$ identically. The lattice therefore contains exactly as many cells of even rank as of odd: 105 of each. The occupied set inherits the balance exactly, at 59 and 59, because it fills whole columns as Section 13.2 establishes.

The factor of two making every capacity even is electron spin. The parity balance is thus a structural consequence of spin degeneracy rather than a numerical accident of the particular ceilings chosen, and it would persist under any extension of the shell or subshell limits.

11. Madelung Ordering and the Janet Correspondence

11.1 The Madelung rule as a linear functional

The Madelung or aufbau rule fills subshells in order of increasing $n + \ell$, ties broken by increasing n . On the lattice, $n + \ell$ is a linear functional, and its level sets are parallel diagonal planes cutting through Λ . The filling sequence of the elements is a sweep of these planes, illustrated in Figure 9(a).

Sorting the 118 occupied cells by $(n + \ell, n, k)$ and reading off atomic number reproduces the true ordering with Kendall $\tau = 0.992$; the 28 inversions are the known lanthanide and actinide irregularities, where measured ground states depart from the idealised rule. The lattice is compatible with Madelung ordering but does not derive it: the rule remains an empirical input, imposed on Λ rather than produced by it.

What the lattice does supply is an explanation for the pattern of period lengths. Computing the volume of each Madelung slice gives 2, 2, 8, 8, 18, 18, 32, 32, 50, 50, 72, 72 — the slices come in exactly equal consecutive pairs, shown in Figure 9(b). This pairing is the reason the periods of the periodic table repeat their lengths, a regularity that the conventional two-dimensional layout must impose by construction and cannot explain. Here it is a counting fact about parallel slices of a three-dimensional structure.

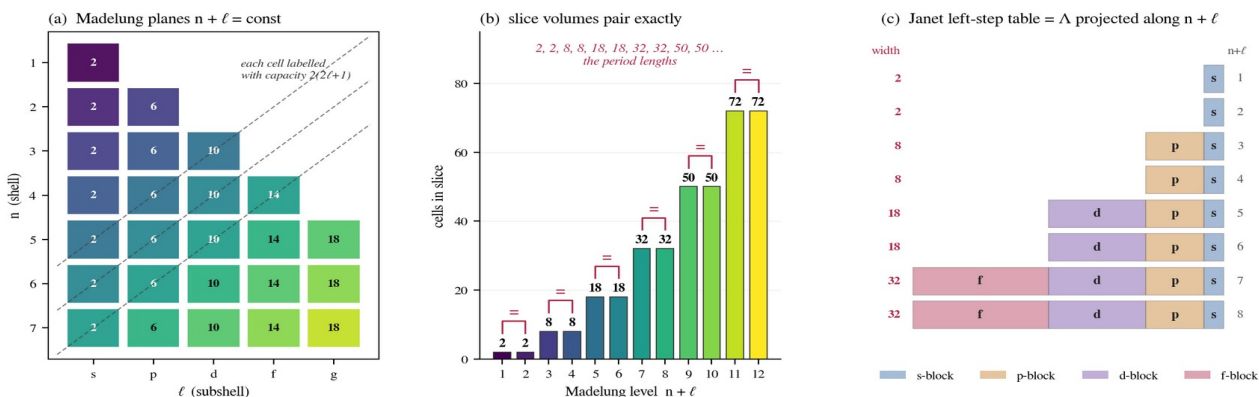


Figure 8. Order-theoretic structure of the lattice. (a) The (n, ℓ) face of Λ , each cell carrying its capacity $2(2\ell+1)$ and shaded by Madelung level; the dashed diagonals are the planes $n + \ell = \text{const}$. (b) Volume of each Madelung slice. The slices pair exactly — 2, 2, 8, 8, 18, 18, 32, 32, 50, 50 — reproducing the observed period lengths of the periodic system. (c) The Janet left-step table, obtained by projecting Λ along the functional $n + \ell$; each row is one Madelung slice and its width is the slice volume from panel (b).

11.2 Correspondence with the Janet left-step table

The projection just described is not a new arrangement. Charles Janet proposed in 1929 a reordering of the periodic table by $n + \ell$ blocks, producing period lengths 2, 2, 8, 8, 18, 18, 32, 32 and placing helium above beryllium rather than above neon. The left-step table has been argued in the philosophy of chemistry to be the most natural periodic form, on the grounds that it follows the quantum-mechanical filling order rather than chemical resemblance.

The relationship to the present framework is exact. Janet's blocks are the level sets of the linear functional $n + \ell$ on Λ , and his table is the two-dimensional projection of the lattice along that functional, as Figure

8(c) shows. This is a correspondence rather than an analogy: the two structures contain the same information, with Janet's form arising from Λ by a specific and identifiable projection.

The claim this supports is modest but real. The left-step table is customarily justified by appeal to the Madelung rule; the lattice shows that it is the shadow of a three-dimensional object, and that the pairing of period lengths which Janet's form exhibits is a volumetric property of that object. Whether this constitutes an improved justification for the left-step arrangement is a question for the philosophy-of-chemistry literature, where such matters are actively debated.

11.3 Chemical groups as fibres of the lattice

The Janet correspondence of Section 11.2 concerns the projection of Λ along $n + \ell$. A second and independent relationship emerges from a different operation on the same lattice: fixing the pair (ℓ, k) and allowing n to vary.

Each such fibre is a line through Λ parallel to the n -axis, and the elements lying on it are precisely a conventional chemical group. Fixing $(\ell, k) = (0, 1)$ gives hydrogen, lithium, sodium, potassium, rubidium, caesium and francium; $(1, 4)$ gives oxygen, sulphur, selenium, tellurium, polonium and livermorium; and so on across the main groups. The columns of the two-dimensional table are one-dimensional fibres of the three-dimensional lattice, and the periodicity that the conventional layout represents by adjacency of columns is here a statement about parallel lines.

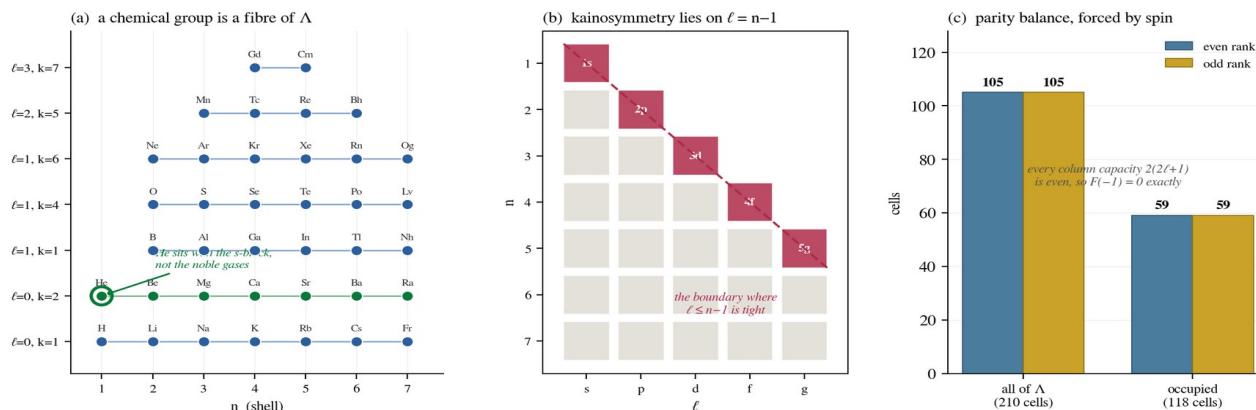


Figure 12. Fibres, the lattice boundary, and parity. (a) Fixing (ℓ, k) and varying n traces a fibre of Λ ; each fibre is a conventional chemical group. The fibre $(\ell, k) = (0, 2)$ contains helium alongside beryllium and the alkaline earths rather than the noble gases. (b) The kainosymmetric subshells 1s, 2p, 3d, 4f and 5g occupy the diagonal $\ell = n - 1$, the boundary at which the constraint $\ell \leq n - 1$ is tight. (c) Parity balance: Λ divides evenly between cells of even and odd rank, and the occupied set inherits the division, a consequence of every column capacity being even.

Seven of the eight main groups are reproduced exactly. The single departure is instructive. Helium has valence configuration $1s^2$, placing it at $(1, 0, 2)$ and hence on the fibre $(\ell, k) = (0, 2)$ together with beryllium, magnesium, calcium and the remaining alkaline earths — not on the fibre $(1, 6)$ occupied by neon, argon and the other noble gases. The lattice separates helium from the group with which the conventional table places it.

This is exactly the placement that the Janet left-step table adopts, and Section 11.2 obtained it by a different route: as a consequence of projecting along $n + \ell$. That two independent operations on Λ — projection along a linear functional, and fibration along the n -axis — converge on the same contested placement is a point in favour of the arrangement, though the question of whether helium is better classified by electron

configuration or by chemical behaviour is one on which the philosophy-of-chemistry literature remains divided, and nothing here settles it.

11.4 The lattice boundary and kainosymmetry

A further correspondence concerns the diagonal $\ell = n - 1$, where the angular-momentum constraint is tight. The cells on this diagonal are 1s, 2p, 3d, 4f and 5g: the subshells that are first of their kind, possessing no radial node.

These subshells are anomalous in well-documented ways. The phenomenon is known as kainosymmetry or primogenic repulsion: a first-of-kind subshell is unusually compact, is not screened by an inner subshell of the same ℓ , and produces the irregularities of the first short period, the anomalous properties of the 3d series relative to 4d and 5d, and part of the divergence of the lanthanides from the actinides. In the conventional table these anomalies appear as scattered exceptions distributed across the layout.

In Λ they are the boundary. The kainosymmetric subshells are exactly the cells where ℓ attains its maximum permitted value for the given n , and their anomalous character corresponds to their lying on the surface of the admissible region rather than in its interior. This is a structural observation rather than an explanation — the lattice does not say why a first-of-kind subshell should be compact — but it does locate a known family of chemical anomalies at a single identifiable feature of the geometry.

12. Why These Coordinates and Not Others

Section 10 established by exhaustive checking that Λ is closed under join and meet. That leaves the more useful question of why, and whether the closure is a coincidence of the particular numbers involved. It is not, and the reason places the lattice within a standard piece of order theory rather than establishing anything new. Stating it correctly also settles the fate of an extension proposed in an earlier draft of this work.

12.1 Λ as an intersection of sublattices

A subset of a product of chains that is closed under componentwise join and meet is a sublattice of that product, and finite intersections of sublattices are sublattices (Birkhoff 1967). The relevant observation is that certain elementary constraint types cut out sublattices of the full box, and that Λ is an intersection of exactly such sets.

Constraints of the forms $x_i \leq x_j$, $x_i \leq c$ and $x_i \geq c$ each define sublattices, as do constraints $x_i \leq f(x_j)$ with f monotone non-decreasing. What these share is that every coordinate appears with the same orientation on both sides: raising the operands raises both the constrained quantity and its bound, and lowering them lowers both. Direct verification on a $6 \times 6 \times 6$ box confirms closure for each such constraint individually and for every pairwise and triple intersection tested.

Λ is built this way. Starting from the box $n \leq 7$, $\ell \leq 4$, $k \leq 18$, which contains 630 cells and is trivially a sublattice, the constraint $\ell \leq n - 1$ reduces it to 450 cells and the constraint $k \leq 2(2\ell + 1)$ to the 210 admissible cells of Λ . Each step intersects with a sublattice, so closure is inherited rather than coincidental. The lower bounds $n \geq 1$, $\ell \geq 0$ and $k \geq 1$ are constants and impose no further condition.

This is a standard argument and no novelty is claimed for it. What is worth recording is that the coordinates of Part I happen to satisfy the condition, which is why the results of Sections 10 and 11 hold.

12.2 Where the hydrogenic labelling fails

An earlier draft observed that the occupancy index k conflates the magnetic quantum number m with the spin projection, and proposed unfolding k into those constituents in the hope of admitting the $SO(4,2)$ representation theory that the three-axis form cannot support. The account just given shows why that cannot work, and locates the obstruction precisely.

The hydrogenic constraint is $|m| \leq \ell$, which is the conjunction of two inequalities: $m \leq \ell$ and $-m \leq \ell$. The first is of the permitted form. The second is not, because the coordinate m enters negated, and orientation is reversed: raising m makes the left-hand side smaller rather than larger. Constraints in which a coordinate appears with mixed orientation do not in general define sublattices, and the same defect appears in familiar examples such as $|x - 2| \leq y$, which fails closure on a 6×6 box.

Building $\Lambda' = \{(n, \ell, m, s)\}$ constraint by constraint shows the failure entering at a single identifiable step. The hydrogenic box is a sublattice; adding $\ell \leq n - 1$ leaves 450 cells and preserves closure; adding $m \leq \ell$ leaves 330 cells and still preserves it; adding $-m \leq \ell$ gives the 210 admissible cells and destroys it. Exhaustive checking of the resulting set returns zero join failures and 4,040 meet failures.

A concrete instance makes the mechanism plain. The cells $(\ell, m) = (3, -3)$ and $(1, 1)$ are both admissible. Their componentwise meet is $(1, -3)$, for which $|-3| > 1$, and the meet leaves the admissible set. Figure 9 contrasts this with the behaviour of k , whose bounds carry no orientation reversal.

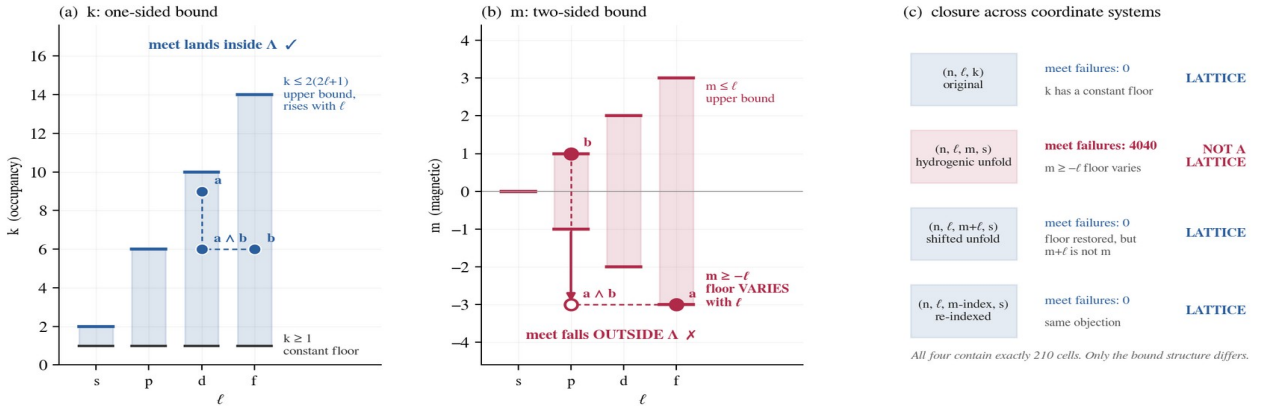


Figure 9. Why the occupancy coordinate is load-bearing. (a) The constraint on k : a ceiling rising with ℓ and a constant floor at $k = 1$, both monotone in the same orientation, so meets remain admissible. (b) The hydrogenic constraint on m : the ceiling $m \leq \ell$ is permitted, but the floor $m \geq -\ell$ places m on the left with reversed orientation. The meet of $(\ell, m) = (3, -3)$ and $(1, 1)$ is $(1, -3)$, which violates $|m| \leq \ell$. (c) Closure across four coordinate systems, each containing exactly 210 cells. Only the signed hydrogenic labelling fails, and it fails on meet alone.

12.3 Why the repair is not a repair

Closure can be restored by substituting $m' = m + \ell$, which converts the pair into $m' \leq 2\ell$ and $m' \geq 0$ — a monotone ceiling and a constant floor, both permitted. Verification confirms that the shifted four-axis set is again closed, and re-indexing the magnetic states from 1 to $2\ell + 1$ achieves the same effect.

But the repair defeats its own purpose. The coordinate m' is not the magnetic quantum number; it is m displaced by an amount that varies across the lattice. The representation theory that motivated the unfolding acts on m , not on m shifted by a position-dependent offset, and so has no more purchase on the shifted lattice than on the original. Closure is recovered precisely by abandoning the object the unfolding was performed to obtain. The proposal is therefore withdrawn: no four-axis refactoring of Λ both

preserves the order structure of Sections 10 and 11 and exposes the hydrogenic labels in a form on which group theory can act.

12.4 What this means for the rest of the paper

The account above clarifies the standing of results elsewhere in this work, in three ways.

It explains rather than merely reports. The facts of Section 10 — that Λ is a bounded lattice, that hydrogen is its unique minimum, that shell capacities appear as slice volumes — follow from the constraints being of a permitted form, rather than resting on exhaustive verification alone. Hydrogen occupies the least position because all three floors are constants and n begins at 1.

It identifies what is load-bearing. The occupancy coordinate k , introduced in Part I as nothing more than the electron count, is the coordinate that makes the structure work: its bounds carry no orientation reversal, and the results of Part II would not survive its replacement by the hydrogenic labels it conflates. This gives independent significance to the finding of Section 8.3, where k was the coordinate carrying information orthogonal to established chemical descriptors. The same coordinate does the work in both the order-theoretic and the empirical results.

And it sets a boundary. Because no admissible refactoring exposes m , the framework cannot be connected to the dynamical symmetry group of the hydrogen atom by any route considered here. Whatever the lattice is, it is not a representation-theoretic object in disguise; it is a combinatorial one, and its correspondences are with other combinatorial descriptions of the periodic system, of which the Janet arrangement of Section 11 is the principal example.

A final observation belongs here, since it concerns where the substance of this work resides. The identities underlying Equation (4) were verified symbolically and hold exactly: the normalisation $\cos^2(\Omega t) + 2[\sin(\Omega t)/\sqrt{2}]^2 = 1$ is identically satisfied, as is $\cos(\Omega t) + \sin(\Omega t) = \sqrt{2} \sin(\Omega t + \pi/4)$. But the summation in Equation (4) carries no element label and its prefactor is element-independent, so all 118 elements share a single envelope. The expression is a correct normalisation statement about Λ rather than a dynamical law distinguishing one element from another; everything element-specific resides in ΔQ and $\sigma^2(t)$, which are geometric quantities. Taken together with the results of this part, the substantive content of the framework is the lattice and its order structure. The field equation of Section 5 is the weakest component, and the present paper is titled for the lattice accordingly.

PART III — THE PERIODIC SYSTEM WITHIN THE LATTICE

Prefatory note. Part II established what kind of object Λ is. This part asks a different question: of the 210 admissible cells, 118 are occupied by known elements — where does that subset sit within the structure, and does its position have any character of its own? The answer is that it does, and that the property in question follows from the physics of how elements are populated rather than from the geometry of the lattice.

13. The Periodic System as an Order Ideal

Sections 10 to 12 concern the lattice as a whole: its closure, its slices, and why the coordinates behave as they do. This section concerns the position within that lattice of the 118 cells that are actually occupied, and reports a property of that subset which does not appear to have been noted.

13.1 The occupied cells are downward closed

An order ideal, or down-set, of a poset is a subset closed downward: if x belongs to it, so does every element below x . Testing this for the occupied cells of Λ — for every occupied x and every admissible y with $y \leq x$ componentwise, is y also occupied? — returns no violations across all 118 cells. The downward closure of the occupied set is exactly the occupied set.

In chemical terms the statement is this. If an element exists whose ground-state valence configuration is (n, ℓ, k) , then every configuration with no larger shell, no larger subshell and no larger occupancy is likewise realised by some element. There are no gaps beneath any occupied cell.

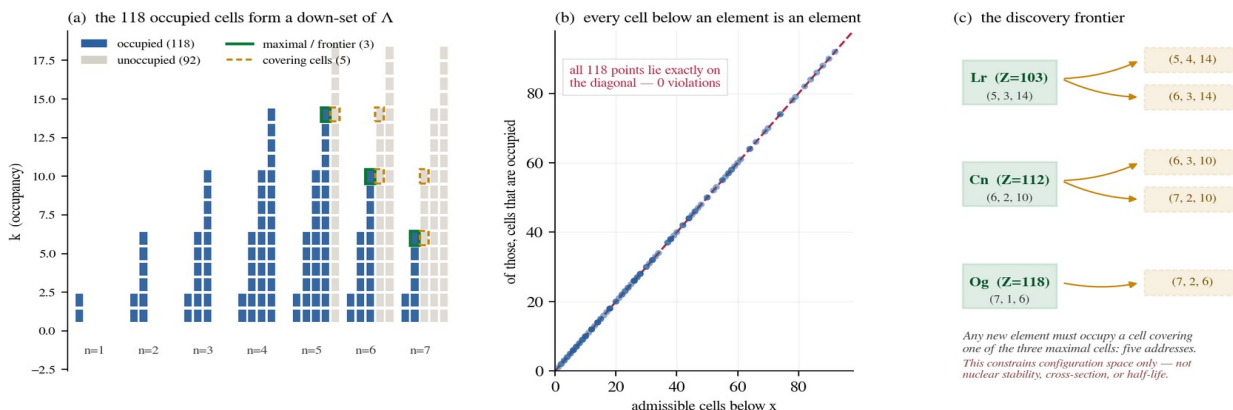


Figure 10. The periodic system as an order ideal of Λ . (a) The 210 admissible cells arranged by shell, with the 118 occupied cells filled. The occupied region is closed downward; the three maximal cells are outlined in green and the five cells covering them are outlined in gold. (b) For each occupied cell x , the number of admissible cells below it against the number of those that are occupied. All 118 points lie on the diagonal, with counts ranging from 1 to 92: the down-set property holds without exception. (c) The discovery frontier. Any element beyond those known must occupy a cell covering one of the three maximal cells, giving five candidate addresses in configuration space.

13.2 The realised ideal is all-or-nothing by column

The down-set property can be sharpened. Group the admissible cells into columns indexed by (n, ℓ) , each of capacity $2(2\ell+1)$, and record for each column how many of its cells are occupied. Of the twenty-five columns of Λ , nineteen are occupied to their full capacity and six are entirely empty. None is partially filled.

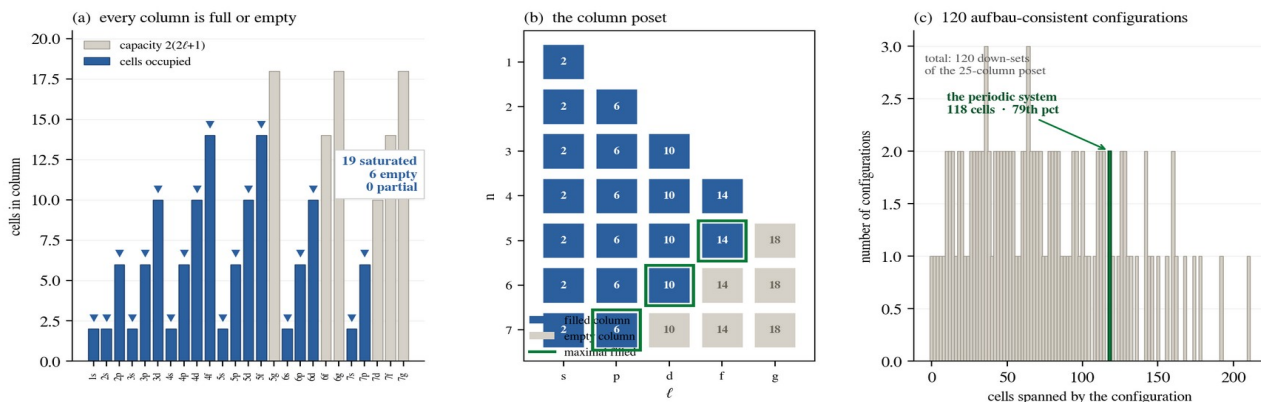


Figure 11. The column structure of the realised ideal. (a) Occupancy against capacity for each of the twenty-five (n , ℓ) columns: nineteen saturated, six empty, none partial. (b) The column poset, with occupied columns shaded and the maximal occupied columns outlined; the realised system is a down-set of this twenty-five element poset. (c) All 120 down-sets of the column poset, by the number of lattice cells each spans. The realised system spans 118 cells and sits at the 79th percentile — among the larger configurations available within a seven-shell lattice, though not the largest.

The reason is again the filling process. A subshell fills to its Pauli capacity before the next begins, so a partially filled column corresponds only to the transient state between two consecutive elements and never to the accumulated set. Configurations such as $3d^6$ exist as the ground state of iron, but by the time the accumulation reaches zinc the $3d$ column is complete, and it remains complete thereafter.

This is a stronger statement than downward closure alone. A general order ideal of Λ may assign any height to each column subject to the closure condition; the realised ideal assigns only zero or capacity. The realised system is therefore determined by a binary choice per column, and the admissible configurations are exactly the down-sets of the twenty-five element column poset. There are 120 of these. The periodic system is one, spanning 118 cells, which places it at the 79th percentile of the 120 by size — among the larger configurations available, though six columns remain unoccupied.

Those six columns are precisely the ones identified in Section 13.4 as the frontier: the g -subshell at $n = 5$, the f - and g -subshells at $n = 6$, and the d -, f - and g -subshells at $n = 7$. In the column description, the growth of the periodic system is the successive filling of whole columns rather than the accumulation of individual cells, and the five covering cells of Section 13.4 are the entry points to three of those six columns.

13.3 Why downward closure holds

The property is not a coincidence, and the reason is worth stating because it distinguishes this result from numerical accidents of the kind discussed in Section 13.5. Elements are populated by aufbau filling: electrons are added one at a time, and the sequence of ground-state configurations traces a monotone path through the lattice. A set generated by a monotone accumulation of this kind is closed downward by construction. The order ideal property is the order-theoretic shadow of the filling process.

This also explains why the property survives the known irregularities. The lanthanide and actinide anomalies, which place 28 inversions in the Madelung ordering of Section 11, alter which cell is reached at which atomic number but do not remove any cell from the accumulated set. Ordering is disturbed; downward closure is not.

13.4 Consequences

Three consequences follow, of decreasing strength.

The first is structural. Λ is distributive, since componentwise minimum and maximum on integer tuples satisfy the distributive laws; direct checking confirms this and identifies 27 join-irreducible cells from which all 210 are generated. By Birkhoff's representation theorem the order ideals of a finite poset themselves form a distributive lattice, so the periodic system corresponds to a single element of the lattice $J(\Lambda)$ of ideals of Λ . The historical growth of the known elements is then a monotone path upward through $J(\Lambda)$, each discovery moving the system to a cell covering its predecessor. This is a description rather than a prediction, but it is an exact one.

The second concerns the frontier. The maximal occupied cells — those with no occupied cell above them — number exactly three: (5, 3, 14) held by lawrencium, (6, 2, 10) held by copernicium, and (7, 1, 6) held by oganesson. Because the occupied set is an ideal, it can grow only by acquiring a cell that covers one of these. There are five such cells: (5, 4, 14), (6, 3, 10), (6, 3, 14), (7, 2, 6) and (7, 2, 10). Any element beyond those presently known must occupy one of these five addresses in configuration space, or else extend the lattice beyond $n = 7$.

The limits of that second consequence require emphasis, and they are the same limits Section 9 places on the reserved cells. The claim constrains valence configuration only. It says nothing about nuclear stability, synthesis cross-sections, half-lives, or whether a nucleus placed at such an address would persist long enough to be observed, all of which are governed by nuclear structure rather than by electron bookkeeping. The statement is that if a new element is characterised, its valence configuration will be found at one of five coordinates — not that any of the five will be reached.

The third is a caution about interpretation. The lattice is a lossless recoding of the element list rather than a compression of it. The joint entropy of the coordinates across the 118 occupied cells is 6.883 bits, exactly $\log_2(118)$, matching the entropy of a flat enumeration; the assignment is injective and no information is discarded or created. What the coordinates buy is organisation — the structures of Sections 10 to 13 — and not a reduction in the descriptive burden.

13.5 A coincidence that is only a coincidence

One numerical observation invites a correspondence that does not hold, and it is recorded here so that others do not pursue it. The lattice contains exactly 210 admissible cells, and 210 is the binomial coefficient $C(10, 4)$, which counts the partitions fitting inside a 4×6 box and hence the elements of a Young lattice of that shape. The coincidence is suggestive.

It does not survive examination. Two graded posets of equal cardinality are isomorphic only if their rank generating functions agree, and these do not. Ranking Λ by $n + \ell + k$ gives level sizes 1, 2, 3, 4, 6, 8, 11, 13, 14, 15, 15, 14, whereas the corresponding Gaussian binomial coefficient gives 1, 1, 2, 3, 5, 6, 9, 10, 13, 14, 16, 16. The two structures differ from the second rank onward. Λ is not a Young lattice, and the equality of cardinalities carries no structural content.

A structural invariant settles the matter independently. The full rank sequence of Λ is 1, 2, 3, 4, 6, 8, 11, 13, 14, 15, 15, 14, 13, 12, 11, 10, 9, 8, 7, 7, 6, 5, 4, 3, 3, 3, 2, 1, which is unimodal with its maximum at rank 11 but is not palindromic. Gaussian binomials are rank-symmetric, as are the Boolean lattices, the divisor lattices and the Young lattices of a box. A lattice whose rank sequence fails to be symmetric cannot be isomorphic to any of them. The asymmetry arises from the interaction of the two ceilings: $\ell \leq n - 1$ truncates the higher subshells at small n while $k \leq 2(2\ell + 1)$ allows the tall f and g columns only where ℓ is already large, so the lattice is broader near its base than near its summit.

14. Relation to Existing Work

Alternative periodic arrangements have a long history — helical, cylindrical, pyramidal and left-step forms among them. These generally reorganise by atomic number and chemical periodicity while preserving a two-dimensional surface. The present construction differs in treating the three quantum numbers as independent geometric axes and, more importantly, in classifying the resulting empty volume into physically distinguished regimes. The mathematical ingredients — bra-ket notation, superposition, the Pauli principle — are entirely standard; what is offered here is a coordinate choice and the structural facts it reveals.

14.1 Order-theoretic treatments of chemical systems

Order theory has an established presence in chemistry, and two lines of work stand close enough to the present one to require explicit comparison.

Leal and Restrepo have developed a formal account of the periodic system as an ordered hypergraph, in which the elements are vertices, chemical similarity classes are hyperedges, and the order relation is induced by measured properties such as atomic number or electronegativity. Their construction and the present one order different objects by different relations: theirs orders elements by properties, the present work orders electron configurations by componentwise comparison of quantum numbers. The two are complementary rather than competing, and a remark of Restrepo's is worth recording — that the several periodic tables in circulation may be understood as projections of a single underlying structure. Section 11.2 arrives at a particular instance of that idea from the opposite direction, obtaining the Janet arrangement as the projection of Λ along $n + \ell$.

Quintero, Brüggemann and Restrepo have separately introduced what they term posetic coordinates, assigning to each object of a poset the triple of cardinalities of its up-set, its down-set and its incomparable set, and mapping those triples into ternary and FOU plots for visualisation. Despite the superficial similarity of vocabulary, the notion of dimension employed there is not the order dimension of Section 10.1: their mappings are visualisation devices reducing a Hasse diagram to a readable plot, whereas order dimension is an intrinsic invariant counting the linear orders needed to reconstruct the relation. Their coordinates describe an object's position within a given poset; the coordinates of the present work define the poset. The two notions are independent, and the results of Section 10.1 neither follow from nor conflict with theirs.

Their machinery is nonetheless applicable to Λ and would answer a question this paper does not address: where the 118 occupied cells sit within the lattice as characterised by their up-set, down-set and incomparability profiles. That is a natural continuation and is noted here as such.

14.2 A note on novelty

The order-theoretic results of Part II use standard techniques throughout: the sublattice argument of Section 12 is elementary, the Sperner property is established by Dilworth's theorem, Birkhoff's representation theorem is invoked as stated, and the dimension argument of Section 10.1 applies the standard example in the textbook manner. What is claimed is the application of these techniques to the configuration lattice, not the techniques themselves.

Whether the particular results are new to the literature has not been established. A limited search located the works discussed above and no closer prior art, but dimension theory alone is the subject of a substantial monograph literature, and the possibility that componentwise-ordered configuration posets have been treated elsewhere cannot be excluded on the evidence gathered here. The results are offered as

correct and checkable rather than as novel, and readers in a position to identify precedent are encouraged to do so.

PART IV — AN EXPLORATORY APPLICATION TO DEUTERIDE HOST SELECTION

Prefatory note. This part is exploratory. It records an attempt to apply the lattice to the selection of metal hosts for deuterium loading, and reports both what the attempt produced and where it failed. The failure is instructive and is set out in full in Sections 18 and 19. Readers interested only in established results should treat Part I as self-contained; Part IV makes no claim to constitute a theory of low-energy nuclear reactions, and its ranking should not be read as a physical prediction.

15. Motivation

The 1989 Fleischmann–Pons experiments used palladium, and palladium has dominated the subsequent literature. The choice was reasonable on practical grounds — Pd absorbs hydrogen readily, is workable, is not toxic, and was available. It was not the product of a search over host materials guided by any structural principle, because no such principle was available.

The adjacency result of Section 7 suggests one possible line of attack. If the late f-block and late 5d metals form a coherent region in configuration space, and palladium sits at the edge of that region rather than within it, then the lattice identifies neighbouring elements that a Pd-centred literature would not naturally have reached. The question this part asks is narrow: does the lattice generate a candidate list worth attention?

Section 8 gives this a firmer footing than geometry alone would. Hydride formation enthalpy — the quantity determining whether a metal loads hydrogen and how tightly it holds it — is predicted from the coordinates ℓ and k with cross-validated $R^2 = 0.51$. Loading behaviour is the precondition for any deuteride experiment, so a framework with demonstrated purchase on hydride thermodynamics has a legitimate claim to relevance in host selection. That claim extends to the chemistry of loading. It does not extend to anything nuclear, and Section 19 is explicit about where the boundary falls.

16. A Heuristic Index

We construct an index intended to capture two intuitions: that an element deep in configuration space presents a more complex electronic environment at interstitial sites, and that higher- ℓ subshells contribute more interstitial density at appreciable filling. The index is

$$H_{\text{cp}}(n, \ell, k) = \sqrt[n^2 + \ell^2 + k^2]{} \cdot \left(k / 2(2\ell+1) \right)^{(\ell/4)} \quad (6)$$

The first factor is the configuration-space distance $|Q_0|$. The second is the fractional subshell filling raised to a power that increases with ℓ , the exponent $\ell/4$ being normalised by the lattice ceiling $\ell_{\text{max}} = 4$.

This expression is a choice, and we flag it as such rather than presenting it as derived. Infinitely many monotonic functions of ℓ and k satisfy the stated intuitions; $\ell/4$ was selected for its normalisation against the lattice ceiling and because it is the simplest such form, not because the framework requires it. A different exponent gives a different index and, as Section 18 shows, a different ordering. The index has no dimensions and no interpretation as a rate, an energy, or a probability. It is a sorting device.

17. The Screen

Two filters are applied to all 118 elements. The first is chemical: the host must form a well-characterised interstitial hydride or deuteride, since a metal that does not absorb deuterium cannot serve as a loading matrix. The second is structural. In an FCC lattice the octahedral interstitial sites are all equivalent by

symmetry and occur six to the unit cell, so three deuterons can occupy translationally equivalent positions. In HCP the octahedral sites are not all equivalent and the geometry depends on the c/a ratio; in DHCP the layer alternation breaks equivalence outright. FCC is therefore taken as the permissive case.

Neither filter derives from the lattice. Both are ordinary crystallography, imported. The lattice contributes only the index by which the surviving candidates are sorted.

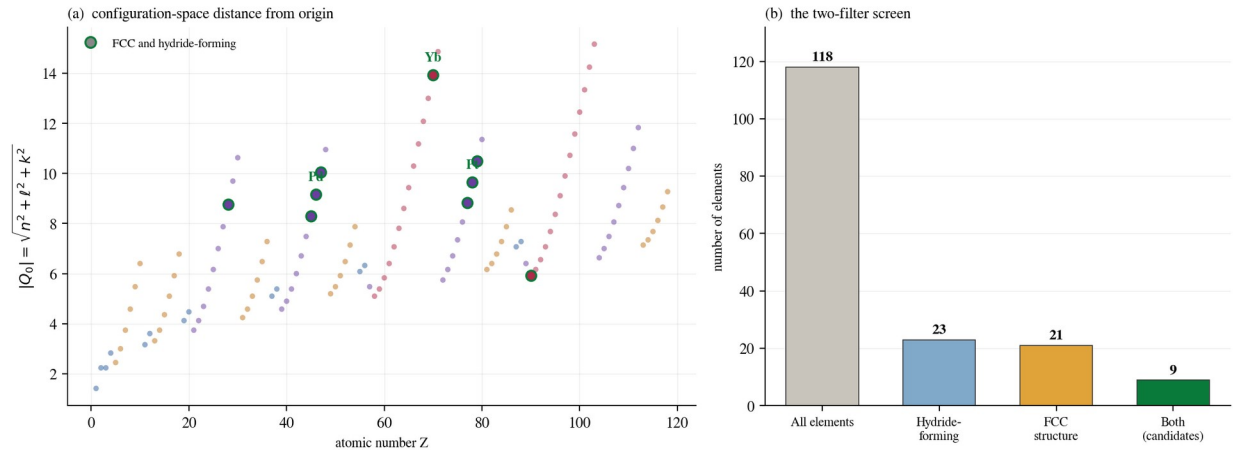


Figure 13. Screening across the elements. (a) Configuration-space distance $|Q_o|$ against atomic number, with elements passing both filters ringed. The three labelled entries are Pd (the historical host), Pt and Yb. (b) The two-filter funnel: 118 elements reduce to nine candidates.

Table 2. Candidates passing both filters, ordered by the heuristic index. Ranking is convention-dependent — see Section 18 and Figure 14.

Rank	El.	(n, ℓ, k)	$ Q_o $	H_{cp}	Class	Note
1	Yb	(4, 3, 13)	13.93	13.18	C	apparently untested
2	Au	(5, 2, 9)	10.49	9.95	B	poor absorber
3	Ag	(4, 2, 9)	10.05	9.53	B	weak absorber
4	Pt	(5, 2, 8)	9.64	8.63	B	well studied
5	Pd	(4, 2, 8)	9.17	8.20	B	historical host
6	Ni	(3, 2, 8)	8.78	7.85	C	well studied
7	Ir	(5, 2, 7)	8.83	7.39	B	little studied
8	Rh	(4, 2, 7)	8.31	6.95	B	little studied
9	Th	(5, 3, 1)	5.92	0.82	B	radioactive

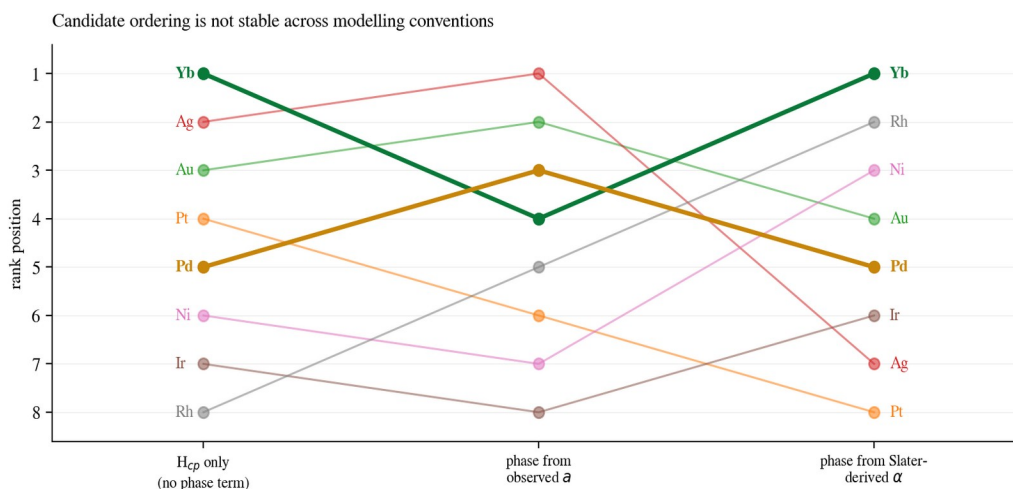
The entry that stands out is ytterbium. It has the largest configuration-space distance of any candidate, it forms YbH_2 and YbH_3 under accessible conditions, it adopts an FCC structure, it is stable and commercially available — and a survey of the deuteride-loading literature turns up no report of its use as a primary host. Whatever the status of the index that ranked it, the observation that a chemically suitable, structurally suitable, commercially available metal appears never to have been tried is independent of the index and stands on its own.

18. Why the Ordering Is Not a Prediction

Table 2 sorts by H_{cp} alone. But the screen involves further modelling choices, and it is instructive to see what happens when they are varied within the range of defensible options.

A phase term may be introduced to represent the geometric relationship between occupied interstitial sites, requiring a lattice parameter. That parameter can be taken from measurement, or estimated from

the lattice via a screening model — Slater's rules, or Clementi–Raimondi values, applied either to the valence orbital or to the bonding orbital. Each of these is a reasonable choice. Each yields a different ordering.



Each column is a defensible choice. The ordering changes between them — so the ordering is a property of the choice, not a prediction of the lattice.

Figure 14. Candidate ordering under three modelling conventions. Ytterbium (green) ranks first under two conventions and fourth under the third; palladium (amber) moves between third and fifth. Because the ordering is not stable across defensible choices, it is a property of the convention adopted rather than a result derived from the lattice.

This is decisive, and we state the conclusion plainly. Yb ranks first under two conventions and fourth under a third; Pt ranks fourth, sixth and eighth. When a result depends on which of several equally justifiable conventions the analyst adopts, that result characterises the analyst's choice and not the structure under study. The ordering in Table 2 is therefore reported as one convention's output, not as a prediction of the framework.

We note also, for the record, that two intermediate steps in the exploratory work were initially presented as derivations and were not. An interaction term of the form $\gamma\hat{Q}\otimes\hat{Q}$ was described as the natural coupling; it is one choice among many, and alternatives give different functional forms downstream. The exponent $\ell/4$ in Equation (6) was described as the unique form satisfying the stated constraints; it is not unique, and it was selected after the index it appears in had already been written down. Both are recorded here as choices.

19. The Dimensional Obstruction

A more fundamental limitation applies regardless of any convention, and no refinement of the screening model addresses it.

H_{cp} is dimensionless, and this is not incidental. Each axis of Λ is a count: n enumerates radial nodes, ℓ enumerates angular nodes, and k enumerates electrons. The lattice is a subset of $\mathbb{N}^3$, and every quantity constructed from its coordinates is therefore a pure number. The obstruction described here is intrinsic to the construction rather than a gap in it, and no refinement of the index can remove it. Deuteron–deuteron fusion, by contrast, proceeds through Coulomb barrier penetration, whose probability goes as $\exp(-G)$ where the Gamow factor G is built from the reduced mass in kilograms, the barrier height in joules, and the

internuclear separation in metres. Converting a dimensionless lattice ratio into a rate requires a dimensional scale, and the framework supplies none.

One might attempt to supply a scale from the physical geometry of the deuterium configuration — an interstitial separation of order 10^{-10} m, say. But the relevant separation for tunnelling is the distance of closest approach, some five orders of magnitude smaller, and any assumed relation between H_{cp} and that separation is precisely the physical content one is trying to derive. The exponential sensitivity of $\exp(-G)$ makes this fatal rather than merely imprecise.

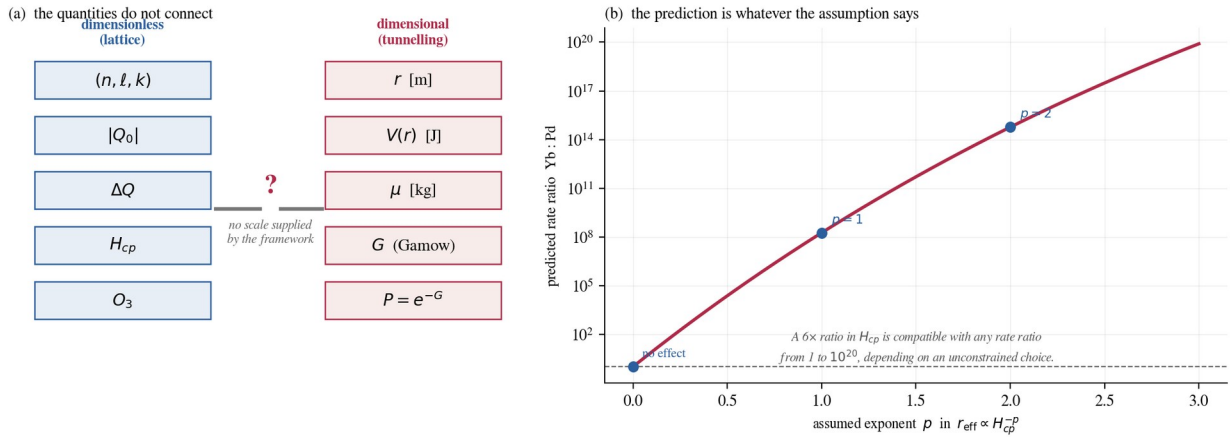


Figure 15. The dimensional obstruction. (a) The lattice produces dimensionless quantities; barrier penetration requires dimensional ones, and the framework provides no bridge. (b) Sensitivity of the predicted Yb : Pd rate ratio to an assumed relation $r_{eff} \propto H_{cp}^p$. Varying the exponent over a modest range moves the predicted ratio across twenty orders of magnitude. The prediction is a restatement of the assumption.

Figure 15(b) makes the point quantitatively. A ratio of roughly six in H_{cp} is compatible with a rate ratio of unity, or of 10^{20} , according to an exponent that nothing constrains. It follows that Part IV cannot yield a falsifiable prediction in the ordinary sense: were ytterbium to show no anomalous behaviour whatever, no quantity in this framework would be contradicted, because none was ever committed to a magnitude.

It might be objected that Section 8 already demonstrated dimensional content in the lattice, and that the obstruction is therefore softer than stated. It is not, and the reason is scale separation. The hydride regression relates coordinates to bonding energies of order 10^{-1} eV over interatomic distances of order 10^{-10} m. Barrier penetration involves energies of order 10^5 eV over separations of order 10^{-15} m. A correlation demonstrated at the first scale carries no implication for the second — that six-order gap in energy and five-order gap in length is the whole reason cold fusion is difficult, and no regression against chemical data can bridge it. Equation (7) predicts how much deuterium a metal will absorb. It says nothing about what happens to the deuterons once absorbed.

There is a further reason for caution. The three-body treatment in the exploratory work was motivated by the lattice having three axes. But (n, ℓ, k) are quantum numbers — a shell index, an angular momentum, a count — and their being three in number is not a statement about physical space. Reasoning from the dimensionality of the coordinate system to the geometry of a physical configuration conflates two distinct senses of "three-dimensional".

20. What Part IV Offers

Stripped of the claims it cannot support, the exploratory work leaves three things.

First, a candidate-generation heuristic. The lattice, combined with two crystallographic filters, reduces 118 elements to nine and directs attention to a region of the periodic system that a palladium-centred literature has not systematically explored. Heuristics of this kind are legitimate and useful provided they are not mistaken for theories.

Second, a specific and checkable observation. Ytterbium satisfies every practical criterion for a deuteride-loading host — FCC structure, established hydride chemistry, stability, availability — and appears absent from the experimental record. That absence is a fact about the literature, not a consequence of the index, and it would remain interesting if Equation (6) were discarded entirely.

Third, a worked negative result. The dimensional obstruction of Section 19 is not peculiar to this framework; it applies to any scheme that attempts to reason from combinatorial or configuration-space quantities to nuclear rates. Documenting where the boundary lies has some value in a field with a long history of frameworks that did not identify their own.

21. Conclusion

Part I offers a coordinate change and the structural facts it makes visible: a classification of the lattice's empty volume into void, reactive and reserved regimes, a census placing 92 cells in the last of these, a sorting of the elements into three geometric classes by the direction of ΔQ , and an adjacency between the late f-block and the late 5d metals that the conventional layout destroys. Tested against hydride formation enthalpy and benchmarked against Pettifor and Miedema, the coordinates are outperformed on the headline task, and the paper reports this. What survives the benchmark is narrow and specific: the occupancy coordinate carries information orthogonal to every established descriptor tested, and improves Pettifor's scale measurably.

Part II contains the results of which this work can be most confident, because they are structural facts about the coordinate system rather than correlations with data, and because each can be checked by direct computation. It opens by settling whether the third coordinate is needed: a standard example S_3 among the admissible cells gives the lattice order dimension exactly three, so no two linear orders can encode its containment relation and the construction of Part I is not an arbitrary choice of three axes over two. The admissible cells form a bounded algebraic lattice whose unique minimum is hydrogen; shell capacities emerge as slice volumes $2n^2$; the lattice is Sperner without being rank-symmetric; its rank generating function vanishes at $q = -1$, an exact parity balance traceable to spin. The Madelung rule is a linear functional whose level sets pair to give the period lengths, and projection along it reproduces Janet's left-step table exactly, while fibration along the n -axis reproduces the chemical groups and places helium among the alkaline earths by a second and independent route. That the kainosymmetric subshells occupy precisely the boundary diagonal $\ell = n - 1$ locates a scattered family of chemical anomalies at a single feature of the geometry. Section 12 explains the closure by a standard sublattice argument rather than a new theorem, shows that the hydrogenic refactoring proposed earlier cannot work, and identifies the occupancy coordinate as the component that makes the structure hold together — the same coordinate that carries the orthogonal chemical information found in Part I.

Part III locates the periodic system within that structure. The 118 occupied cells form an order ideal of Λ , a property inherited from the monotone character of aufbau filling, which places the periodic system as one point in the lattice of ideals and reduces the frontier of possible growth to five addresses in configuration space. The ideal is all-or-nothing by column, making the realised system one of 120 admissible configurations rather than one of a far larger family. These are descriptions rather than predictions, but

they are exact ones, and they distinguish what the lattice contributes from what the physics of filling contributes.

Read together, the four parts describe a combinatorial object with genuine order-theoretic structure and a definite relationship to existing periodic formalisms, modest empirical purchase on hydride chemistry through one of its coordinates, and no reach into nuclear physics. The field equation of Section 5 is, on the evidence assembled here, the least substantial component of the framework. The lattice is the contribution, and the paper is named for it.

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